Machine Learning Project

Comparative Evaluation of Machine Learning Classifiers for Alzheimer's Disease Detection on a Synthetic Health Dataset

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Abstract—Alzheimer's disease (AD) represents a major global health challenge, with prevalence projected to triple by 2050. This study evaluates the performance of eleven Machine Learning models for the binary classification of Alzheimer's disease onset, using a synthetic dataset containing general health information for 2,149 patients across 35 variables. The primary research objective is to identify which classification model best supports AD detection by maximizing sensitivity. AdaBoost emerged as the optimal algorithm, achieving a cross-validated sensitivity of 0.932 and test accuracy of 0.926, demonstrating strong generalization with minimal overfitting (train-test gap: 0.032). SHAP-based explainability analysis identified FunctionalAssessment as the dominant predictive factor, followed by ADL (Activities of Daily Living) and MemoryComplaints, providing clinically interpretable insights into model predictions.

1. INTRODUCTION

Alzheimer's disease (AD) is a form of dementia, a neurological condition characterized by progressive impairment of memory and cognitive abilities, as well as changes in behavior and personality. AD is considered the most prevalent form of dementia among the elderly worldwide, with an incidence of approximately 5% among individuals aged over 65. Consistent with this epidemiological profile, the dataset used in this study includes subjects aged 65 to 90 years, with a median age of 75.

1.1. Disease Detection

Because no specific pre-symptomatic screening tests are currently available, diagnosis is not established through targeted examinations conducted before symptom onset or at very early stages. A diagnosis of probable Alzheimer's disease can therefore be made only on the basis of the patient's clinical history and tests assessing alterations in cognitive functioning. Early detection remains one of the main current research priorities.

Overall, Alzheimer's disease onset is generally considered to be influenced by a combination of genetic, environmental, and lifestyle factors. However, partly due to the sporadic occurrence of the disease across individuals, a clear hereditary pattern has not been demonstrated, despite the existence of genetic factors that can increase the risk of developing the condition.

1.2. Significance of Studying the Disease

According to a study published in *BMC Psychiatry*, the total number of Alzheimer's disease and other dementia cases more than doubled between 1990 and 2021, increasing from approximately 19.8 million to about 52.6 million. Moreover, projections reported by Alzheimer's Disease International and *The Lancet Public Health Forecasts* estimate that dementia cases will rise to roughly 153 million by 2050, effectively tripling current levels. In this scenario, timely identification becomes crucial to enable earlier implementation of disease management strategies.

Within this perspective, the present study evaluates the performance of eleven Machine Learning models for the binary classification of disease onset, using the presence/absence and intensity of specific individual risk factors. Early and prompt diagnosis may also facilitate earlier access to emerging experimental treatments or palliative options, such as those investigated in the clinical study led by Professor Giacomo Koch at Fondazione Santa Lucia IRCCS, supported by the Italian Ministry of Health and the Italian Ministry of University and Research.

2. RESEARCH QUESTIONS

1. *Which classification model best supports Alzheimer's disease detection by maximizing sensitivity?*
2. *How can the trade-off between predictive performance and model interpretability be addressed?*

This study aims to use ML techniques to maximize sensitivity for AD detection while addressing the interpretability–performance trade-off.

3. DATASET DESCRIPTION

This study uses a synthetic dataset containing general health information for 2,149 patients, each identified by a unique `PatientID`. The data were retrieved from Kaggle and include 35 variables grouped into eight thematic domains. The target variable is `Diagnosis` (0/1), indicating the absence/presence of a prior Alzheimer's diagnosis. One column (`DoctorInCharge`) contains confidential information and is constant ("XXXConfid") for each patient.

- **Dataset URL**: https://www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset

3.1. Variables by Domain

The dataset variables are organized across the following domains:

3.1.1. *Patient Identifier*

- `PatientID`: Unique identifier

3.1.2. *Confidential Information*

- `DoctorInCharge`: Confidential; value "XXXConfid" for each patient

This report was prepared for the Machine Learning Project for Data Science course at Università degli Studi di Milano Bicocca. The project evaluates eleven machine learning models for binary classification of Alzheimer's disease onset using a synthetic health dataset.

3.1.3. Demographics

- `Age`: 65–90 years
- `Gender`: 0=male, 1=female
- `Ethnicity`: 0=Caucasian, 1=African American, 2=Asian, 3=Other
- `EducationLevel`: ordinal 0–4 (higher = higher education)

3.1.4. Lifestyle Factors

- `BMI`: continuous
- `Smoking`: 0=smoker, 1=non-smoker
- `AlcoholConsumption`: weekly units 0–20
- `PhysicalActivity`: weekly hours 1–10

3.1.5. Medical History

- `FamilyHistoryAlzheimers`: 0=no, 1=yes
- `CardiovascularDisease`: 0=no, 1=yes
- `Diabetes`: 0=no, 1=yes
- `Depression`: 0=no, 1=yes
- `HeadInjury`: 0=no, 1=yes
- `Hypertension`: 0=no, 1=yes

3.1.6. Clinical Measurements

- `SystolicBP`: mmHg 90–180
- `DiastolicBP`: mmHg 60–120
- `CholesterolTotal`: mg/dL 150–300
- `CholesterolLDL`: mg/dL 50–200
- `CholesterolHDL`: mg/dL 20–100
- `CholesterolTriglycerides`: mg/dL 50–400

3.1.7. Cognitive and Functional Assessments

- `MMSE`: Mini-Mental State Examination, 0–30 (higher = better cognition)
- `FunctionalAssessment`: 0–10 (lower = worse functional status)
- `MemoryComplaints`: 0=no, 1=yes
- `BehavioralProblems`: 0=no, 1=yes
- `ADL`: Activities of Daily Living, 0–10 (lower = worse daily function)

3.1.8. Symptoms

- `Confusion`: 0=no, 1=yes
- `Disorientation`: 0=no, 1=yes
- `PersonalityChanges`: 0=no, 1=yes
- `DifficultyCompletingTasks`: 0=no, 1=yes
- `Forgetfulness`: 0=no, 1=yes

3.1.9. Target Variable

- `Diagnosis`: 0=no Alzheimer's, 1=Alzheimer's

4. DATA PRE-PROCESSING

4.1. Workflow Reproducibility

To ensure full reproducibility of the workflow and all analyses, a fixed random seed (123) was set and explicitly used in every step involving stochasticity (e.g., partitioning and sampling), so that repeated executions yield consistent results.

4.2. Data Loading and Preliminary Checks

The dataset was imported into a KNIME workflow. `PatientID` was first removed from the original dataset, as it is not a predictive variable. We then checked for duplicate records; no duplicate rows were detected, hence no deduplication was required.

4.3. Encoding of Categorical Variables

Categorical variables were handled through dedicated transformations. In particular, binary quantitative variables were converted from numeric values to string labels to explicitly reflect their boolean nature. For Ethnicity and Education Level, dummification was applied. It is worth noting that this dummification (KNIME One to Many) was not implemented as a strict one-hot encoding. The potential collinearity introduced by this approach is addressed during the feature selection phase, where redundant features can be identified and removed. Furthermore, for Decision Trees and MLP models, sensitivity to collinearity is negligible.

4.4. Dataset Size and Target Distribution

The final dataset used in the workflow consists of 2,149 observations and 39 columns. The target variable (`Diagnosis`) includes 1,389 observations with no diagnosed Alzheimer's (0) and 760 observations with diagnosed Alzheimer's (1). This class imbalance was considered in subsequent split and re-balancing decisions.

4.5. Train/Test Partitioning

The dataset was split into training and test sets using the Table Partitioner node, with a 70% train / 30% test split. To preserve the target distribution across subsets, stratified sampling was performed on `Diagnosis`. After splitting, the training set contains 972 instances of class 0 and 532 instances of class 1.

4.6. Handling Class Imbalance

Given the imbalance described in Section 4.4, downsampling was applied to the majority class (0) to match the size of the minority class in the training set. This was performed using approximate sampling with a static seed, for improved practicality in memory management and to control the stochastic component of the analysis.

4.7. Feature Standardization

To make numerical variables comparable and facilitate model training, features were standardized using Z-score normalization. Standardization parameters (mean and variance) were estimated on the training set and then applied to normalize the test set using the same values.

4.8. Feature Selection

A feature selection pipeline was implemented using a supervised Decision Tree with `Diagnosis` as the target. The Gini Index was adopted as the impurity measure, with a constraint of at least 2 records per node. The procedure was embedded

in a loop that evaluates feature combinations and selects the subset maximizing accuracy.

The best configuration achieved an accuracy of 0.909, selecting 32 features out of 37. The variables automatically excluded by the feature-selection loop were: `FamilyHistoryAlzheimers` (consistent with findings from the Telethon Association; link to website), `CardiovascularDisease`, `DiastolicBP`, `DifficultyCompletingTask`, `Ethnicity_Afroamerican`, and `Education_higher`.

Notably, the exclusion of the last two dummy variables (`Ethnicity_Afroamerican` and `Education_higher`) addresses the collinearity issue introduced by the non-strict one-hot encoding applied during the categorical variable transformation phase. By removing one category from each dummified variable, the feature selection process effectively establishes a proper reference category, eliminating multicollinearity. The same feature subset was then applied to the test data.

5. DATA EXPLORATION

5.1. Target Balancing

After applying under-sampling to the training set, the target variable becomes effectively balanced: class 0 includes 534 observations, whereas class 1 includes 532.

5.2. Correlation Analysis and VIF of Quantitative Variables

We examined correlations among quantitative variables to identify potentially problematic relationships. Overall, correlation coefficients are close to zero, indicating a low degree of linear association among numerical features. Additionally, the Variance Inflation Factor (VIF) analysis yielded values around 1, confirming the complete absence of multicollinearity and validating the suitability of the quantitative feature set for modeling.

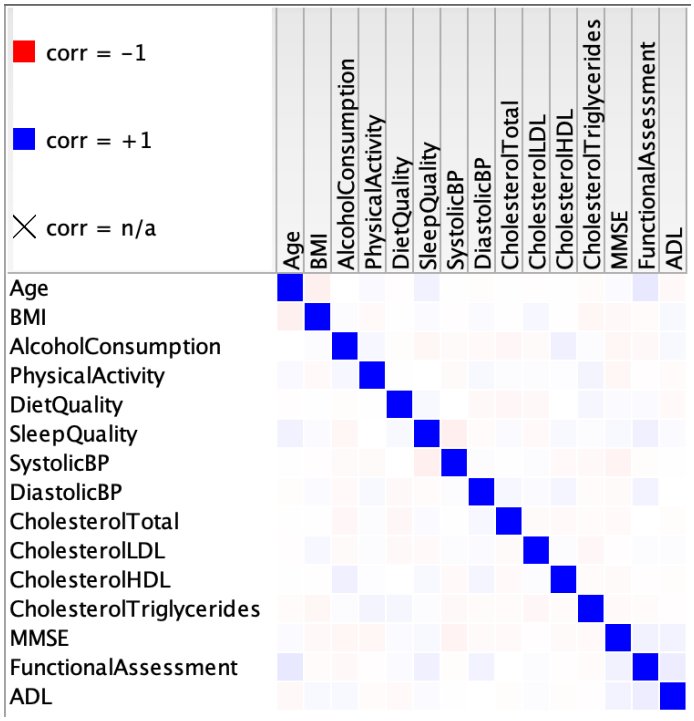


Figure 1. Correlation matrix of quantitative variables in the dataset.

5.3. Zero-variance Features

We checked for features with zero variance, i.e., constant variables that typically provide no informative signal. No such features were found; therefore, no removals were necessary.

5.4. Gender Distribution in the Training Set

The balanced training set includes both males and females in a largely homogeneous manner (543 males and 523 females). This distribution helps exclude the possibility that model performance is driven by a gender imbalance.

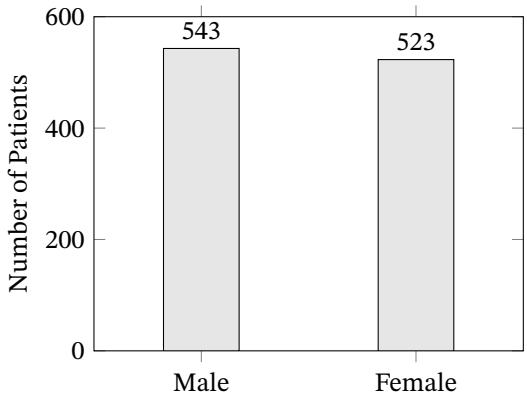


Figure 2. Gender distribution in the balanced training set.

5.5. Age Distribution

Regarding age, the median in the training set is 75 years, with a minimum of 60 and a maximum of 90 years.

6. MODEL TRAINING AND MODEL SELECTION CRITERION

To assess model generalization, we evaluated models using both a holdout strategy (train/test split) and k-fold cross-validation. The train–test partitioning was primarily used to check for potential overfitting, whereas 10-fold cross-validation was used to compute cross-validated metrics, specifically accuracy and sensitivity. For selecting the best-performing model, we relied on cross-validated sensitivity.

In the context of screening for neurodegenerative conditions such as Alzheimer's disease, optimizing sensitivity is a priority over maximizing overall accuracy. High sensitivity ensures that most truly positive cases are correctly identified, accepting a trade-off in specificity that results in more false positives. This methodological choice reflects the precautionary principle typical of medical screening, where minimizing false negatives is the primary clinical objective.

7. MODEL DEVELOPMENT AND EVALUATION

7.1. Models

To enable an exhaustive comparative evaluation, we benchmarked eleven Machine Learning classifiers spanning distinct methodological families: a tree-based model (Decision Tree),

ensemble methods (Random Forest, Bagging, AdaBoost, Gradient Boosting), a linear model (Logistic Regression), a kernel method (SVM), a neural network (MLP), and probabilistic/graphical models (Naive Bayes, Bayesian Network, Tree-Augmented Naive Bayes—TAN). This heterogeneous portfolio supports a systematic comparison of complementary learning paradigms for Alzheimer's detection on the considered dataset.

7.1.1. Decision Tree

The Decision Tree was configured with a confidence factor of 0.25, controlling post-growth pruning aggressiveness: relatively low values promote simpler, more generalizable trees, mitigating overfitting. The `minNumObj` parameter was set to 2, allowing the model to capture patterns even within small subgroups. Pruning was enabled to prevent excessive growth on the training data, balancing interpretability and generalization.

7.1.2. Random Forest

Random Forest was configured with 100 trees, using the Gini Index as the split criterion. The `minNumObj` parameter was set to 2, while `limitnumberofLevels` was disabled. This allows potentially complex individual trees, while the ensemble reduces overfitting through bagging and random feature selection at each split. Stochasticity was controlled via a fixed seed.

7.1.3. AdaBoost

Real AdaBoost was configured with 200 iterations, defining the maximum number of sequential weak learners. Each weak learner is a decision stump (depth-1 decision tree). The shrinkage parameter was set to 0.5, introducing a learning rate that regularizes the contribution of each added classifier; values below 1 slow learning, increasing robustness and reducing overfitting, potentially at the cost of requiring more iterations to converge.

7.1.4. Gradient Boosting

Gradient Boosting was configured with 200 sequential models, where each new tree corrects residual errors from previous learners via gradient descent on the loss function. Individual trees were constrained to a maximum depth of four levels, balancing the expressiveness of each weak learner with the need to keep learners simple and complementary within the ensemble.

7.1.5. Bagging

Bagging trains multiple versions of a base classifier on bootstrap samples of the training set and aggregates predictions to obtain the final output. The base learner was a REP-Tree (Reduced Error Pruning Tree) with: maximum depth $M = 2$, minimum variance for split $V = 0.001$, minimum instances per node $N = 3$, and seed $= 1$ for reproducibility. The Bagging configuration used 100 iterations, corresponding to the number of base classifiers trained, with a bag size of 100% of the training set (sampling with replacement).

7.1.6. Logistic Regression

Binomial Logistic Regression was configured using a regularized maximum-likelihood approach. The ridge parameter was set to 0.001, introducing an L2 penalty on model coefficients to reduce overfitting and improve numerical stability. Correlation analysis among quantitative variables did not indicate collinearity issues; therefore, automatic variable selection via L1 (LASSO) penalization was not required. The maximum number of iterations was set to 100 to ensure convergence.

7.1.7. Support Vector Machine (SVM)

The SVM identifies the optimal separating hyperplane by maximizing the margin between classes, determined by support vectors near the decision boundary. The overlapping penalty (C) was set to 1.0, balancing margin maximization with training-set misclassification. A polynomial kernel with power $= 1.0$ was used, corresponding to a linear kernel. Although strict linear separability may appear unlikely given task complexity, more flexible kernels (quadratic and cubic polynomial, hyperbolic tangent, and RBF) were also tested, but achieved lower performance. The gamma and bias parameters were kept at their default values (1.0), providing a balanced configuration without introducing major distortions in feature space.

7.1.8. Multi-Layer Perceptron (MLP)

The MLP was trained using Resilient Backpropagation (RProp). Network architecture was selected through manual parameter tuning; the final configuration includes two hidden layers with 10 neurons each. Training was run for up to 100 epochs, with seed $= 123$ to ensure reproducibility. The KNIME RProp MLP Learner does not explicitly report the activation function in the available documentation or configuration options; it is reasonable to assume a sigmoid or tanh activation, both commonly used in classic MLP architectures.

7.1.9. Naive Bayes

Naive Bayes assumes conditional independence among predictors given the class, simplifying posterior probability estimation. The model was configured with `useSupervisedDiscretization = true`, enabling supervised discretization of continuous variables. Other options were left at default values; in particular, `useKernelEstimator` was disabled, which is computationally efficient while maintaining good predictive performance. Despite the independence assumption often being violated, Naive Bayes performed well in this context, offering a favorable balance between computational efficiency and predictive effectiveness.

7.1.10. Bayesian Network

Unlike Naive Bayes, a Bayesian Network learns conditional dependencies among variables, representing them as a directed acyclic graph (DAG). Conditional probabilities were estimated using SimpleEstimator with alpha $= 1.0$ (Laplace smoothing to avoid zero probabilities). Network structure was learned via TabuSearch, configured with a maximum of 5 parents per node,

balancing representational power and computational complexity. The TabuSet size was set to 10, with a single optimization run. Candidate structures were evaluated using ScoringBayes.

7.1.11. Tree-Augmented Naive Bayes (TAN)

TAN partially relaxes the Naive Bayes independence assumption by allowing each predictor to have at most one additional predictor as a parent (besides the target), yielding a tree-structured dependency model. Conditional probabilities were estimated with SimpleEstimator using a small alpha ≈ 0.1, applying minimal smoothing close to pure maximum-likelihood estimation. Tree structure learning used the TAN algorithm with BAYES scoring, which identifies the most relevant dependencies between predictors relative to the target by building a maximum spanning tree based on conditional mutual information. This approach maintains the controlled computational complexity of Naive Bayes while capturing the most significant inter-feature dependencies.

8. RESULTS

Sensitivity	Algorithm	Data
0.932	AdaBoost	K-FOLD
0.924	Naive Bayes	K-FOLD
0.924	Bayesian Network	K-FOLD
0.921	Bagging	K-FOLD
0.920	TAN	K-FOLD
0.913	Decision Tree	K-FOLD
0.911	Gradient Boosting	K-FOLD
0.884	Random Forest	K-FOLD
0.818	MLP	K-FOLD
0.787	Logistic Regression	K-FOLD
0.679	SVM	K-FOLD

Table 1. Cross-validated sensitivity by algorithm.

8.1. Detailed Model Analysis

8.1.1. Decision Tree

The Decision Tree exhibited highly stable cross-validation performance, with zero variance across folds. The 10-fold cross-validated mean accuracy was 0.944 (interval 0.918–0.959), supporting model robustness. Accuracy on the training set was 0.963, consistent with cross-validated results. On the test set, accuracy was 0.932, corresponding to a limited train–test gap of 0.031, which suggests good generalization and no clear overfitting. The 10-fold cross-validated sensitivity was 0.913, indicating strong ability to correctly identify positive cases.

8.1.2. Random Forest

Random Forest showed stable cross-validation behavior, again with zero variance across folds. The 10-fold cross-validated mean accuracy was 0.934 (interval 0.906–0.959), indicating consistent performance across data splits. Training accuracy reached 0.988, notably higher than cross-validated estimates. Test accuracy was 0.932, yielding a train–test gap of 0.056, suggesting some tendency toward overfitting. The cross-validated sensitivity was 0.884, lower than the best-performing algorithms, pointing to a more limited ability to identify Alzheimer's positive cases.

8.1.3. AdaBoost

AdaBoost achieved excellent and stable cross-validation performance, with zero variance across folds. The 10-fold cross-validated mean accuracy was 0.936 (interval 0.895–0.965), confirming robustness. Training accuracy was 0.947, closely aligned with cross-validated performance, while test accuracy was 0.926, resulting in a minimal train–test gap of 0.021 and indicating strong generalization. The cross-validated sensitivity was 0.932, the highest among the evaluated algorithms, highlighting AdaBoost's superiority in correctly detecting positive Alzheimer's cases.

8.1.4. Gradient Boosting

Gradient Boosting produced stable cross-validation results with zero variance across folds. The 10-fold cross-validated mean accuracy was 0.935 (interval 0.895–0.959). On the training set, the model achieved perfect accuracy (1.000), substantially exceeding cross-validated performance. Test accuracy decreased to 0.921, producing a sizeable train–test gap of 0.079, a clear indicator of overfitting. The cross-validated sensitivity was 0.911, competitive but below AdaBoost.

8.1.5. Bagging

Bagging delivered excellent and stable cross-validation performance, with zero variance across folds. The 10-fold cross-validated mean accuracy was 0.949 (interval 0.912–0.965), indicating high robustness. Training accuracy was 0.961, in line with cross-validated estimates, and test accuracy was 0.936, with a small train–test gap of 0.025, suggesting strong generalization without meaningful overfitting. The cross-validated sensitivity was 0.921: competitive, though slightly lower than AdaBoost. Despite higher cross-validated accuracy, the lower sensitivity indicates a somewhat reduced ability to detect positive Alzheimer's cases.

8.1.6. Logistic Regression

Logistic Regression performed worse than most other algorithms. The 10-fold cross-validated mean accuracy was 0.821, with minimal fold variance (0.001) and an interval of 0.795–0.865, indicating stable but lower overall performance. Training accuracy was 0.836, and test accuracy was 0.825, with a minimal train–test gap of 0.011, suggesting consistent generalization despite limited absolute performance. The cross-validated sensitivity was 0.787, the lowest among the tested models (excluding SVM), indicating insufficient capacity to capture the complexity and non-linear relationships in the data.

8.1.7. Support Vector Machine (SVM)

The SVM showed the lowest performance overall. The 10-fold cross-validated mean accuracy was 0.753, with minimal fold variance (0.001) but unsatisfactory accuracy levels; the interval was 0.708–0.807. Training accuracy was 0.753, and the train–test gap was 0.021 (i.e., 0.732 on the test set), confirming that

the model does not overfit but rather fails to learn effective patterns from the data. The cross-validated sensitivity was 0.679, the lowest among all algorithms—critical in a clinical context where missed positive cases may delay timely therapeutic intervention.

8.1.8. Multi-Layer Perceptron (MLP)

The MLP achieved intermediate performance with limited fold variability. The 10-fold cross-validated mean accuracy was 0.863, with minimal fold variance (0.001) and an interval of 0.807–0.906. Training accuracy was 0.973, while test accuracy dropped to 0.833, yielding a large train–test gap of 0.140, the widest among the evaluated models and indicative of strong overfitting. The cross-validated sensitivity was 0.818, placing the MLP below several alternative approaches; combined with its overfitting behavior, this suggests it is not optimal in this setting.

8.1.9. Naive Bayes

Naive Bayes achieved excellent performance with optimal stability, evidenced by zero variance across cross-validation folds. The 10-fold cross-validated mean accuracy was 0.943 (interval 0.912–0.965). Training accuracy was 0.945, consistent with cross-validated results, and test accuracy was 0.932, with a minimal train–test gap of 0.013, indicating excellent generalization without overfitting. The cross-validated sensitivity was 0.924, the second-highest after AdaBoost, showing strong detection capability despite the conditional-independence assumption.

8.1.10. Bayesian Network

Bayesian Network produced results identical to Naive Bayes, again with zero fold variance. The 10-fold cross-validated mean accuracy was 0.943 (interval 0.912–0.965), with training accuracy 0.945 and test accuracy 0.932, yielding the same train–test gap of 0.013. The cross-validated sensitivity was 0.924, matching Naive Bayes and second only to AdaBoost. Despite explicitly modeling dependencies via a graph structure, performance remained equivalent, suggesting that the captured dependencies did not provide a meaningful advantage over the simpler classifier.

8.1.11. Tree-Augmented Naive Bayes (TAN)

TAN also achieved excellent and stable performance, with zero variance across folds. The 10-fold cross-validated mean accuracy was 0.946 (interval 0.901–0.965). Training accuracy was 0.954, and test accuracy was 0.930, with a train–test gap of 0.024, supporting good generalization. The cross-validated sensitivity was 0.920, slightly below AdaBoost. TAN represents an intermediate compromise between Naive Bayes and Bayesian Network: it relaxes conditional independence by allowing dependencies between pairs of features via a tree structure. Despite very high accuracy, its slightly lower sensitivity places it in an intermediate position among the Bayesian classifiers.

9. ROC CURVE ANALYSIS

ROC curves are a core tool for evaluating binary classifiers, as they visualize the relationship between the True Positive Rate (sensitivity) on the y-axis and the False Positive Rate as the decision threshold varies. An ideal model would rapidly approach the upper-left corner (0,1), achieving perfect sensitivity (100% true positives) with no false positives, while the diagonal black line represents a random classifier. The Area Under the Curve (AUC) summarizes ROC performance as a single value: AUC = 1.0 indicates a perfect classifier, whereas AUC = 0.5 corresponds to chance-level performance. AUC can also be interpreted as the probability that the model assigns a higher score to a randomly chosen positive sample than to a randomly chosen negative sample.

Algorithm	AUC
Bagging	0.925
Decision Tree	0.922
Naive Bayes	0.922
Bayesian Network	0.922
Random Forest	0.920
TAN	0.920
AdaBoost	0.920
Gradient Boosting	0.915
MLP	0.819
Logistic Regression	0.819
SVM	0.720

Across the evaluated models, ROC performance forms two main clusters (see Figure 3 for ROC curve comparison at the end of the document). The higher-performing group includes Bagging (AUC = 0.925); Decision Tree, Bayesian Network, and Naive Bayes (all with AUC ≈ 0.922); AdaBoost, TAN, and Random Forest (all with AUC = 0.920); and Gradient Boosting (AUC = 0.915). Their curves approach the upper-left corner quickly, reaching high TPR at low FPR, which indicates strong discriminative ability. Lower-performing algorithms are MLP (AUC = 0.819), Logistic Regression (AUC = 0.819), and SVM (AUC = 0.720), whose curves remain closer to the random baseline.

A relevant insight emerges when comparing AUC and sensitivity. Although Bagging does not achieve the highest cross-validated sensitivity (0.921), it attains the largest AUC, suggesting a particularly favorable balance between sensitivity and specificity across the full range of possible thresholds—useful when the decision threshold may need adjustment. Conversely, AdaBoost, with higher sensitivity (0.932) and a competitive AUC (0.920), remains the preferred choice when maximizing the identification of positive cases is the primary objective.

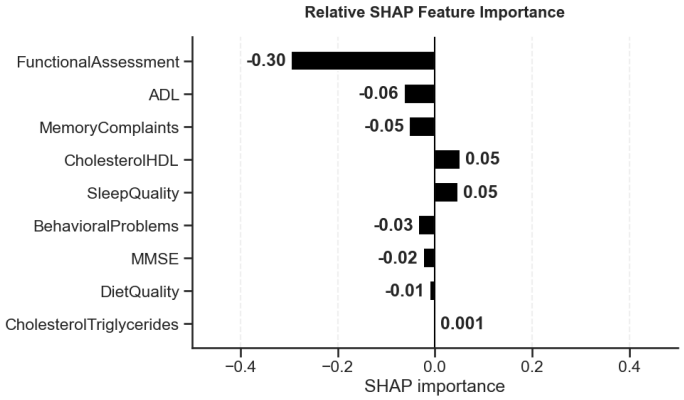
10. HYPERPARAMETER TUNING

After selecting AdaBoost as the optimal algorithm, we performed a hyperparameter optimization phase to verify whether model performance could be further improved. Tuning was implemented using sensitivity as the optimization metric. We optimized (i) the number of iterations (50–300, step 10) and (ii) the shrinkage parameter (0.1–1.0, step 0.1) using a brute-force search strategy. The optimal configuration was `numIterations = 60` and `shrinkage = 0.1`, achieving a cross-validated sensitivity of 0.953 on the training set. However, test-set evaluation revealed an unexpected pattern: the tuned model achieved a sensitivity of 0.899, equal to the default model. The tuned configuration exhibited a larger train–test sensitivity gap

(0.054) than the default model (0.032, from 0.932 to 0.900). We interpreted this as a symptom of overfitting induced by the tuning procedure. Based on these considerations, we adopted AdaBoost with default parameters.

11. SHAP EXTENDED GLOBAL EXPLAINABILITY

To better understand AdaBoost behavior and identify the most influential predictors, we conducted an eXplainable AI analysis using SHAP (SHapley Additive exPlanations). SHAP provides a theoretically grounded measure of feature importance based on cooperative game theory, quantifying each variable's contribution to the prediction for individual observations. In this work, local explanations were extended to a global perspective by aggregating SHAP values over a random sample of the test set.



The analysis was performed on 287 observations: for each instance, SHAP values were computed for every feature, representing the marginal contribution of that variable to the prediction relative to the model's baseline. The local-to-global extension was obtained by summing SHAP values for each feature across all sampled observations. This aggregation preserves SHAP signs, enabling the identification of both magnitude and direction of feature contributions.

Results indicate that `FunctionalAssessment` is the dominant feature, with a negative importance of −0.30, substantially larger than all other variables. The negative contribution implies that lower functional assessment values increase Alzheimer's risk, confirming the central role of functional decline in diagnosis. The next most influential predictors are `ADL` (−0.06) and `MemoryComplaints` (−0.05), indicating that reduced daily living abilities and the presence of memory complaints increase disease risk. On the protective side, `CholesterolHDL` (0.05) and `SleepQuality` (0.05) show positive contributions, suggesting that higher HDL cholesterol and better sleep quality reduce Alzheimer's risk.

12. CONCLUSIONS

This study evaluated 11 Machine Learning algorithms for Alzheimer's disease classification. AdaBoost emerged as the optimal approach, achieving a sensitivity of 0.932 and a test-set accuracy of 0.926, with strong generalization. The global SHAP analysis over 287 observations identified `FunctionalAssessment` as the dominant predictive factor, followed by `ADL` and `MemoryComplaints`. The main limitations concern dataset size and provenance, as the data are synthetic. Future work includes validation on external datasets, extending the task to multiclass classification to distinguish disease stages, and integration into clinical workflows to assess the practical value of decision support.

BIBLIOGRAPHY

- Alzheimer's Disease International. (2022, January 7). *New study predicts the number of people living with Alzheimer's disease to triple by 2050*. https://www.alzint.org/news-events/news/new-data-predicts-the-number-of-people-living-with-alzheimers-disease-to-triple-by-2050/
- El Kharoua, R. *Alzheimer's Disease Dataset* [Data set]. Kaggle. https://www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset
- Fondazione Telethon. (2024, September 20). *Malattia di Alzheimer, le risposte alle domande più comuni*. https://www.fondazionetelethon.it/storie-e-news/news/dalla-ricerca/malattia-di-alzheimer-le-risposte-alle-domande-piu-comuni/
- Fondazione Telethon. *Malattia di Alzheimer*. https://www.fondazionetelethon.it/cosa-facciamo/ricerca/malattie-studiate/malattia-di-alzheimer/
- GBD 2019 Dementia Forecasting Collaborators. (2022). Estimation of the global prevalence of dementia in 2019 and forecasted prevalence in 2050: An analysis for the Global Burden of Disease Study 2019. *The Lancet Public Health*, 7(2), e105–e125. https://doi.org/10.1016/S2468-2667(21)00249-8
- Koch, G., Casula, E. P., Bonnì, S., Borghi, I., Assogna, M., Di Lorenzo, F., Esposito, R., Maiella, M., D'Acunto, A., Ferraresi, M., Mencarelli, L., Pezzopane, V., Motta, C., Santarnecchi, E., Bozzali, M., & Martorana, A. (2025). Effects of 52 weeks of precuneus rTMS in Alzheimer's disease patients: A randomized trial. *Alzheimer's Research & Therapy*, 17(1), Article 69. https://doi.org/10.1186/s13195-025-01709-7
- Yu, D.-T., Li, R.-X., Sun, J.-R., Rong, X.-W., Guo, X.-G., & Zhu, G.-D. (2025). Global mortality, prevalence and disability-adjusted life years of Alzheimer's disease and other dementias in adults aged 60 years or older, and the impact of the COVID-19 pandemic: A comprehensive analysis for the global burden of disease 2021. *BMC Psychiatry*, 25, Article 503. https://doi.org/10.1186/s12888-025-06661-2

APPENDIX A

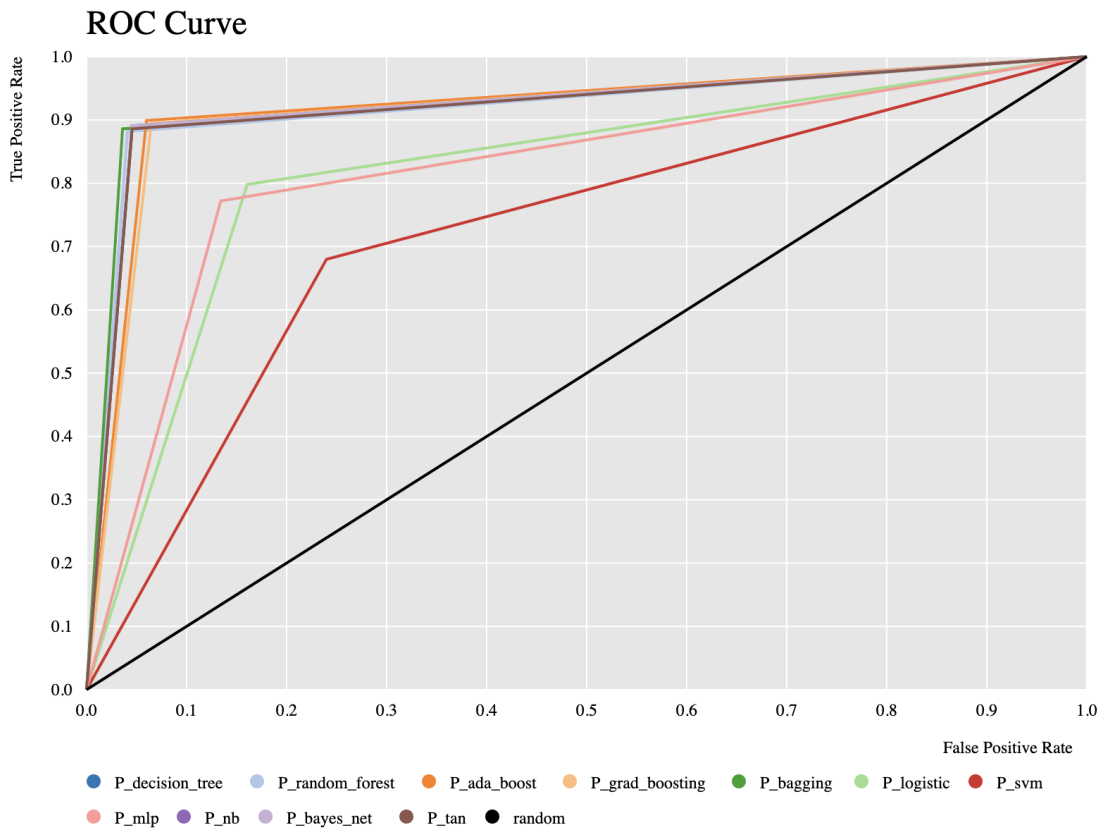


Figure 3. ROC curves for all evaluated classifiers. The diagonal line represents random performance (AUC = 0.5).